

Revision Letter

SIBGRAPI 2026 — Paper ID 173

Visual Similarity Is Not Enough: Domain-Adapted Synthetic Data for Maritime Vessel Detection

Dear Program Chairs and Reviewers,

We thank the three reviewers for their careful and constructive reviews. We have addressed every point raised. Beyond textual clarifications, the revision includes *new experimental evidence* produced specifically for this rebuttal: (i) a multi-seed replication of the pre-training epoch ablation at the endpoints requested by Reviewer 2; (ii) a new similarity-vs-structure diagnostic over the full curated pool that addresses the top- k curation question without confounds; (iii) the zero-shot evaluation of the A' seq. arm requested by Reviewer 3; and (iv) formal seed-paired significance tests replacing the informal “non-overlapping ranges” criterion. All new scripts and per-seed result files have been committed to the anonymised repository cited in the paper.

A PDF with all changes highlighted (generated with `latexdiff`) is provided alongside this letter. Section, table, and figure numbers below refer to the revised paper.

Reviewer 1

R1.1 — The title claim should be softened or scoped.

Response. We kept the title but scoped the claim precisely: the abstract, Section III, and the conclusion now state that it is *global, whole-image* visual similarity that fails to predict transferability. More importantly, the claim is now backed by a mechanism rather than a caveat: a new diagnostic (Section IV-A) recomputes the curation score for all 27,796 curated images and shows the similarity ranking to be nearly orthogonal to the structural axes that cause the negative transfer (see R2.W4 below).

Changes. *Abstract; Section III (dataset paragraph); Section IV-A; Conclusion.*

R1.2 — The joint-rand ablation deserves open discussion.

Response. Agreed. The revised Section IV-B reports the out-of-domain comparison openly: sea-aware random placement is not merely comparable but *consistently better* out of domain (+1.91 pp over A' joint on SMD, all three seeds, seed-paired $p = 0.005$). We now frame anchoring as an engineering choice rather than the mechanism: we recommend in-place anchoring when auditability and the absence of placement heuristics matter (every synthetic position inherits a real observation; no sea-region model or inpainting required), and random placement when cross-domain generalisation is the priority.

Changes. *Section IV-B (final paragraph); Section IV-C.*

R1.3 — Formal statistical testing.

Response. All headline comparisons are now assessed with two-sided seed-paired t -tests (seeds matched across arms), replacing the informal criterion. The central claims hold: A' joint vs. B2, +1.00 mAP50 in every seed, $p = 0.024$; mAP50-95 +1.50, $p < 0.001$; Recall +2.2, $p = 0.013$; negative transfer of A and B, both $p \leq 0.005$. Equivalences are also confirmed as non-differences (A vs. B, $p = 0.71$; B2-long vs. B2, $p = 1.0$; joint-rand vs. joint in-domain, $p = 0.90$). Where a comparison does *not* survive testing we now say so explicitly (see R2.W2). A footnote explains why the paired t -test is the appropriate small-sample choice (with $n=3$ an exact sign-permutation test cannot attain $p < 0.25$).

Changes. *Section III-C (protocol + footnote); Sections IV-A, IV-B, IV-C throughout.*

R1.4 — Is crop scale conditioned on vertical position (perspective)?

Response. No, and the revised text says so explicitly. Section III-B now specifies the full A' joint-rand procedure: real vessels removed by fast-marching inpainting (Telea, radius 3 px; new reference [25]); sea region defined by an adaptive horizon (top of the highest real bounding box), a floor at 35% of the image height, and an HSV refinement that excludes vertical obstacles; rejection sampling with $\geq 75\%$ of the box inside the region and pairwise IoU < 0.2 ; crop sizes drawn from the empirical width–height distribution of all real boxes, *independently of vertical position* — perspective is deliberately not modelled. That the unmodelled variant matches (and out of domain exceeds) the anchored one reinforces our interpretation that the gain stems from crop diversity and the training regime, not geometric realism.

Changes. *Section III-B (new paragraph).*

R1.5 — At which resolution is the size stratification computed?

Response. We resolved this by computational reproduction rather than recollection: we re-implemented the size-stratified evaluation with `pycocotools`, per seed, under both candidate conventions. The convention that reproduces the published Table III is COCO thresholds in *native image pixels* (1920×1080); the alternative (640×640 canvas) diverges by an order of magnitude. Both conventions are now declared where used: Table I’s 71.6% small-object statistic uses the 640 canvas (matching the network input), Table III uses native pixels; the captions state this. Table III itself was regenerated from the committed script with per-seed values (minor changes ≤ 0.01 , same directions), and the per-seed artifacts are in the repository.

Changes. *Captions of Tables I and III; Table III values; repository (`ap_by_size.py` + per-seed JSON).*

Reviewer 2

W1 — Seed-level stability; sea-region procedure.

Response. Seed-level orderings are now reported: $A' \text{ joint} > B2$ in 3/3 seeds (likewise mAP50-95 and Recall); $A \text{ and } B < B2$ in 3/3; $A' \text{ seq.} < B2$ in 3/3. The sea-region and sampling procedure is fully specified (see R1.4). The anchored-vs-random comparison is discussed openly, including the out-of-domain result favouring the random variant (see R1.2).

Changes. *Sections III-B, IV-A, IV-B, IV-C.*

W2 — Statistical support, in particular for the small-object claim.

Response. The reviewer was right on the exact metric: with the regenerated per-seed values, AR_{small} improves in all three seeds ($0.390 \rightarrow 0.411$, +2.1 points) but does not reach significance at $n=3$ ($p = 0.13$). The revision reports this honestly: improvements are consistent in direction across *all* six size-stratified metrics and all seeds, significant for medium and large objects ($p < 0.05$), with the largest point estimates — but no significance claim — for small objects. The abstract, results, discussion, and conclusion were all reworded accordingly.

Changes. *Abstract; Section IV-B; Table III; Discussion (“Operational relevance”); Conclusion.*

W3 — The epoch ablation uses a single seed.

Response. We ran the replication the reviewer requested at the endpoints. The 10-epoch point now has three seeds: mAP50 0.8210 ± 0.0036 , below the COCO baseline in all three seeds; the 100-epoch endpoint (three seeds, 0.7936 ± 0.0049) is significantly below the 10-epoch point (seed-paired $p = 0.028$), confirming under replication that degradation grows with exposure. Figure 4 was regenerated: filled markers with error bars at 0, 10, and 100 epochs; open markers explicitly mark the remaining single-seed points (20, 50). Incidentally, the multi-seed endpoint strengthens the trend: the previous single-seed 100-epoch value (0.8006) happened to be the most optimistic of the three seeds.

Changes. *Figure 4 (figure, caption); Section IV-A.*

W4 — Would a stricter top- k similarity curation change the outcome?

Response. Rather than train on one specific top- k (which would confound curation with pre-training-set size), we answer the question for the entire family of global-similarity criteria. We recomputed the curation score for all 27,796 curated images under the original protocol and profiled similarity deciles against the structural variables. The ranking is nearly orthogonal to them: Spearman $\rho = -0.152$ against vessel area, and object density is constant at one vessel per image, so similarity has no selective power on that axis at all. Even the top-5% most similar images have a median vessel area of 29% — versus 0.1% operationally — so any stricter threshold merely climbs the same ranking towards the same structural profile. No global-similarity threshold can close the gap; this is now stated with the numbers in Section IV-A, and the former “left to future work” sentence was replaced accordingly. The diagnostic script and artifacts are in the repository.

Changes. *Section III (dataset paragraph); Section IV-A (new passage); repository (clip_decile_profile.py + artifacts).*

W5 — The 50/50 joint protocol is ambiguous.

Response. Now fully specified: 13 variations per real training image (1,348 real $\rightarrow$ 17,524 synthetic); in the balanced joint regime the real set is repeated $13\times$, so one A' joint epoch contains 35,048 images (17,524 real + 17,524 synthetic, 50/50 by construction). B2-long matches this volume with no synthetic data.

Changes. *Section III-B; Section III-C (B2-long definition).*

Reviewer 3

R3.1 — What exactly is A' seq., and why 100 epochs?

Response. The definition is now explicit: a two-stage pipeline — 100 epochs of pre-training on the synthetic set, then 300 epochs of fine-tuning on real CITRA-3D-Real only; unlike A' joint, the two sources are never mixed (and unlike B2-long, which contains no synthetic data). Extending the synthetic stage beyond 100 epochs is counter-indicated by the monotonic degradation of Figure 4; the results text now makes this connection. We also qualified the -1.31 figure: below the baseline in all three seeds but not significant at $n=3$ ($p = 0.14$).

Changes. *Section III-C; Section IV-B.*

R3.2 — A' seq. is missing from the zero-shot table.

Response. Added: A' seq. reaches 0.6026 ± 0.0373 mAP50 on SMD (+1.58 vs. B2, the largest seed variance in the table, hence statistically indistinguishable from B2). This is informative: the arm that degrades

in-domain does *not* degrade out of domain, indicating that the damage of the sequential regime is specific to in-domain fine-tuning rather than a general corruption of features. The interpretation is now in Section IV-C.

Changes. *Table IV; Section IV-C.*

R3.3 — Notation in the experimental-setup list.

Response. Fixed as suggested: A' joint (COCO $\rightarrow$ real+synthetic combined 50/50 in a single run).

Changes. *Section III-C.*

R3.4 — Table II is too wide.

Response. Converted to percentages (e.g., 84.51 ± 0.33), as suggested; the caption declares the unit and Δ in percentage points.

Changes. *Table II.*

Corrections made on our own initiative

While reproducing every number for this revision from committed scripts, we found and fixed two inaccuracies; we report them here for full transparency.

1. **Public-set vessel area:** $\sim 54\% \rightarrow 39\%$ (median). The previously reported $\sim 54\%$ could not be reproduced. The value measured on the actual pre-training set (27,796 images, committed script) is a median of 38.9% (mean $\sim 41\%$). Figure 1, Table I, the abstract, and the related-work discussion were updated, and the derived scale ratio was corrected from $\sim 500\times$ to $\sim 400\times$. The paper’s argument is unaffected — if anything the corrected measurement is conservative, and it now has full provenance.
2. **Table IV, B2-long:** $0.5755 \pm 0.0131 \rightarrow 0.5706 \pm 0.0063$. We discovered that the previously reported value included one seed evaluated with an intermediate checkpoint (its training run finished later the same day and overwrote the file). Re-evaluating all three seeds with the final checkpoints yields the corrected value; two of the three seeds reproduce the original numbers exactly. The derived deltas in the text were updated ($-1.1 \rightarrow -1.6$ pp vs. B2; A' joint gap $+2.47 \rightarrow +2.96$ pp). The qualitative conclusion (mild out-of-domain harm from real-image oversampling) is unchanged, indeed slightly strengthened.

Table III was likewise regenerated from the committed evaluation script with per-seed values (changes ≤ 0.01 , same directions throughout); the repository now contains every script and per-seed artifact needed to reproduce Tables II–IV and Figures 1 and 4.

We believe the revision addresses all concerns and thank the reviewers again for comments that measurably improved the paper’s rigour.

The Authors